

Q1

Software Quality Assurance is a planned and systematic set of activities used to ensure that software meets defined quality standards and customer expectations.

Software Quality Assurance is a structured, proactive framework aimed at upholding established quality standards throughout software creation. By focusing on process optimization and defect prevention, SQA spans every phase of the Software Development Life Cycle (SDLC) ranging from initial requirements analysis to post-launch maintenance.

High software quality is significant as poor quality leads to losses in money, security, trust from customers, and dissatisfaction from users. It is crucial to maintain the high quality of software as it is critical to protect the brand and profitability of a business. Critical issues such as failing to pay for something or application downtime are responsible for damaging the customer trust, reducing the user retention due to one experience, and generating monetary losses. Moreover, high quality of software helps businesses avoid any severe legal repercussions in controlled industries by eliminating security issues. Early detection of bugs and errors prevents any quality debt and high-cost fixes.

Q2

Quality Assurance is process-oriented and focuses on preventing defects by improving development processes, standards, audits, and reviews. On the other hand, Quality Control is product-oriented and focuses on identifying defects in the software product through inspection and review. QA is mainly proactive, while QC is mainly reactive.

Quality Control (QC) is product-oriented and involves inspecting and reviewing the software to identify defects. Testing is execution-oriented and focuses on executing test cases and performing activities such as functional, regression, integration, and exploratory testing to find bugs and verify software behavior. Therefore, testing is a specific activity used within the broader quality-control process.

Quality Assurance or QA is a way of working that tries to stop problems before they happen. It does this by setting rule checking work and looking at how software's built. On the other hand software testing is all about finding problems that already exist. It is more about trying out the software and seeing if it works the way it should. In the end QA is about stopping issues from appearing while testing is, about finding and fixing problems that're already there.

Q3

AI is changing software testing by moving it from old ways of using fixed scripts to more smart and flexible methods. Of scripts that stop working when the user interface changes AI based testing can understand what needs to be done change when needed and get better with each time it is used. AI now helps with all parts of the software development life cycle by helping with understanding what needs to be done making test cases looking at code running tests and dealing with problems. Also AI makes automation that can fix itself by changing parts and gives smart reports by explaining what went wrong grouping similar problems and finding why they happened.

Q4

A QA Engineer's key responsibilities span the full testing life cycle, blending traditional testing tasks with AI feature validation. They write comprehensive test cases covering all potential feature scenarios, generate required test data, and execute tests to confirm that software behaves as expected. When issues arise, QA Engineers analyze defects to identify root causes and write detailed bug reports for developers. Additionally, they build and maintain automated scripts to streamline repetitive testing tasks and document the overall SQA process. For AI-powered applications, QA Engineers specifically validate features—such as chatbots—to evaluate prompt understanding, ensure contextual accuracy, eliminate hallucinations, and prevent severe security risks like sensitive data leaks.

For day to day task A QA Engineer typical day revolves around making sure the software actually works properly before real users get their hands on it. First, they spend time reading through product requirements and feature ideas to figure out what needs to be tested and catch confusing details early. Next, they write down step-by-step test cases covering different ways a feature could break, and they set up fake test data, like dummy user accounts, to run those checks. During the actual testing phase, QA Engineers manually click through the app or run automated scripts to see if everything behaves right. When something goes wrong or breaks, they investigate what happened and write up simple, detailed bug reports so the developers know what to fix.

Software Test Engineers have to make sure that an application works perfectly fine before any real users access it. First of all, engineers prepare test cases for all scenarios how a certain functionality may be used by someone. Besides, they create dummy test data, such as user accounts, for conducting tests. Once testing begins, Software Test Engineers perform all test cases prepared earlier with an aim to check if the application functions correctly. When there is some problem with the application, Software Test Engineers determine the reason and write clear bug reports in order to inform developers about necessary improvements. In order not to click repetitive actions, Software Test Engineers develop automated test scripts. Finally, if there are

any intelligent functionalities in the application, such as a chatbot, then software test engineers also test this functionality to check that it responds correctly and does not disclose confidential information.

For day to day task Software Test Engineers involve a lot of things in order to ensure that quality of an application is checked all the time. they carry out requirement analysis by studying feature specification to know what to test is about . After that they proceed with writing test cases which involve the process of planning out how to test an application using either manual or AI means, then comes test data generation, a process through which they obtain the necessary test data. In test execution, they carry out tests of the application build for any errors. Once errors occur, they deal with bug report writing in order to log defects. To cut down on tedious activities, they engage themselves in automation coding by generating automated test scripts. Lastly, they undertake documentation of their testing efforts.

AI Quality Engineers have a specialized responsibility in the validation and security of intelligent systems during their development phase. They validate AI applications in terms of their correctness and relevance of output responses and make sure that there are no hallucinations, which means they check that the system does not produce any fake information. In order to confirm that the AI is correctly understanding the user's request, AI Quality Engineers validate the context and prompt. Furthermore, there are security validations performed by AI Quality Engineers in order to protect the system from leaking out any personal information like passwords, OTPs, CVVs, and ATM PINs. For better efficiency, AI Quality Engineers make use of prompt engineering principles using frameworks like RTCCCF.

For day to day work AI Quality Engineers assessing, optimizing, and securing AI-powered applications by prompt engineering, model validation, and safety testing. They dedicate their time to designing and optimizing prompts with the help of frameworks such as RTCCCF for achieving greater accuracy and response reliability. they perform tests on test sets to measure the relevance of responses, hallucination rate, and ability of the AI to maintain context while handling multi-turn conversations. They carry out adversarial testing like red-teaming and prompt injection tests to make sure that the AI-powered application does not expose any sensitive information such as password, PIN, and user-specific details. Whenever defects are noticed, the AI Quality Engineers record detailed defect reports to help the Developers optimize prompts or training data. Ultimately, they document the performance metrics of the models and work closely with Developers and Product Managers on human-in-the-loop assessments.

Q5

